



Davao del Norte State College

INSTITUTE OF COMPUTING

Bachelor of Science in Information Technology



CAPSTONE — **PROJECT** GUIDELINES

A Capstone Project Guidelines for Bachelor of Science in Information Technology students' manuscript. This will serve as a reference for drafting the project's Introduction to Methodologies manuscript.

THIS CAPSTONE GUIDELINES IS INTENDED ONLY FOR THE STUDENTS USE WHO ARE OFFICIALLY ENROLLED IN THE IT 323 CAPSTONE PROJECT 1 COURSE (AY 2022 - 2023). NOT FOR DISTRIBUTION OUTSIDE ITS INTENDED USE

Revised 2023

JOMARIE DAVE JAVA - Faculty

LAARNI C. IBAÑEZ - Faculty

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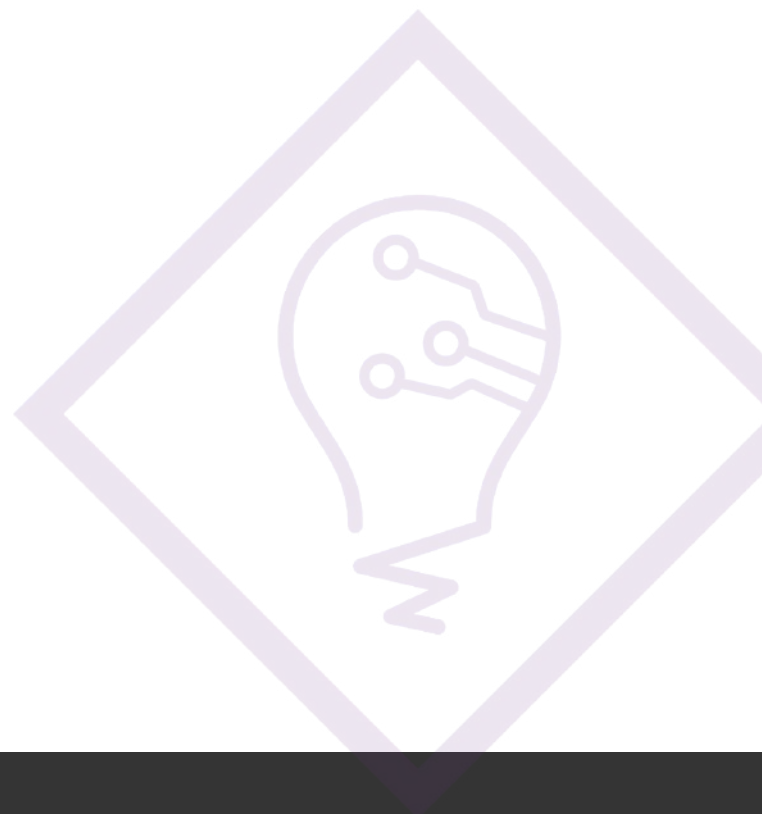
MARK VAN M. BULADACO, MIT - IC Dean



Major Course Output: As evidence of attaining the mentioned course learning outcomes, students are required to do and submit the following during the indicated the term. The rubrics for these outputs are provided.

Course Learning Outcome	Program Outcomes													Assessment Tools	Time Frame
	01	02	03	04	05	06	07	08	09	10	11	12	13		
CLO 1	D	D	D	D		E	E	D	D		D	D	D	Title Defense	Midterm
CLO 2								D		D					
CLO 3	D	D	D	D	E	D	D	D	D		D	D	D	Outline Defense	Final term

The Capstone project is a three-semester undertaking required to complete the BSIT program.





INTRODUCTION

IT 323 Capstone Project 1 is a 3-unit course in the 3rd year 2nd semester of the BSIT curriculum based on CMO 25 series of 2015. The pre-requisite to enroll in this course is 3rd year standing.

Course Description: This course focuses on the infrastructure, application and processes involved in implementing a computing solution to a problem.

Course Learning Outcomes: Upon completion of this course, the students are expected to:

CLO1: Apply the appropriate Systems Analysis and Design tools, methods, models /techniques in analyzing business processes to identify problems that can be solved using IT solutions by conceptualizing capstone project proposals.

CLO2: Present and defend the conceptualized capstone project proposals to various audiences.

CLO3. Develop the accepted capstone project proposal by prototyping the initial system user interface design.



PROJECT TEAM COMPOSITION

The Capstone Project team comprises of three (3) to four (4) members regardless of the number of BSIT 3rd-year enrollees. The following are the roles that the proponents/researchers should play:

1. **Project Manager** - The person with the authority to manage a Capstone Project. This includes leading the planning and the development of all Capstone Project deliverables. The project manager is responsible for the budget, work plan, and all Project Management Procedures. He is responsible for the success of the entire activity.
2. **Systems Analyst and Technical Writer** - The person who checks that all parts of the system are coordinated. A person who finalizes the Research / Capstone Project study document, both the system and the Research / Capstone Project manuscript.
3. **System Designer** - The person who makes sure that the database design is complete and robust. The person who masters the system's network design and prepares the User-Interface design (Forms/ Screenshots/ Storyboard).
4. **System Developers** - These peoples are the ones who design, write, and test computer programs.



CAPSTONE PROJECT ADVISORY COMMITTEE

The proponents will be able to choose their adviser and panel committee members. However, if the Dean of the Institute approves, the Capstone project instructor/s may designate a predetermined set of panel committee members to ensure they are fit for the respective projects assigned. The Project Advisory Committee is composed of the following:

ONE (1) PROJECT ADVISER

1. He or She must be a faculty of the Institute of Computing.
2. Ensures that the study proposed by the students conforms to the standard of the College and has an immediate or potential impact on the research thrust of the College
3. Guides the team in writing the technical paper and system development in the Title proposal stage, outline stage, pre-deployment stage, and deployment stage.
4. Meets the team regularly (at least twice a month, NOTE: the team must seek proper appointment) to answer questions and help resolve impasses and conflicts.
5. Recommends the team for Title Defense, Outline Defense, Pre-Deployment Defense, and Final Defense.

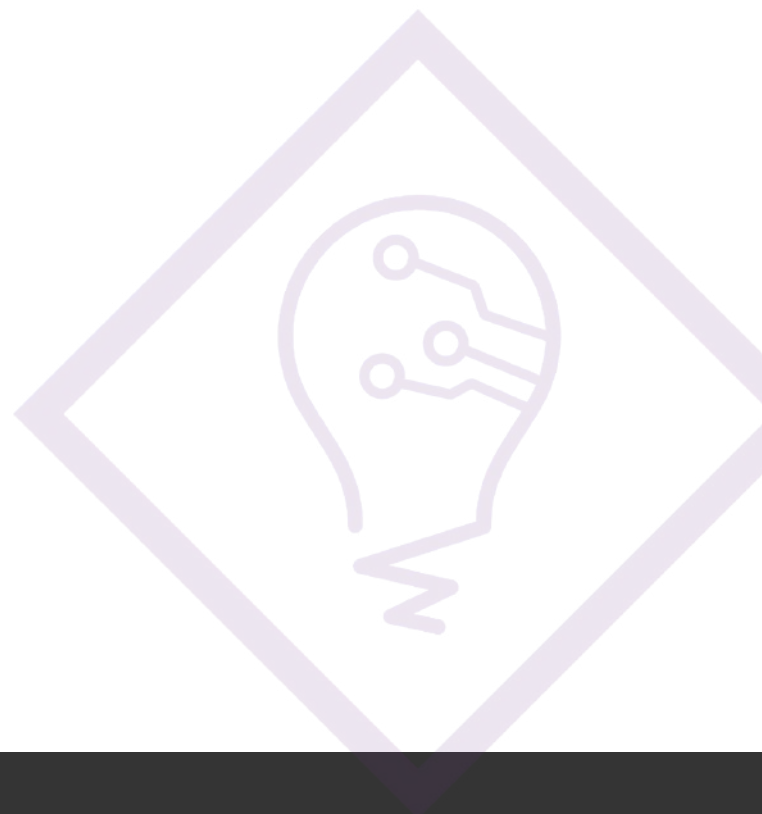


ONE (1) PANEL CHAIRMAN

1. He or She must be a faculty of the Institute of Computing.
2. Brief the team about the Title Defense, Outline Defense, Pre-Deployment Defense, and Final Defense.
3. Issue the verdict. The verdict is a unanimous decision among the three members of the panel. Once issued, it is final and irrevocable.

TWO (2) PANEL MEMBERS

1. He or She must be a faculty of the Institute of Computing.
2. Evaluate the technical paper and the system development.
3. Recommend a verdict.
4. Listen and consider the request of the project adviser and the team members.





PROJECT THEME

Capstones are more concerned about system development. The proponents must identify what a specific organization or agency requires and then determine what product would meet that demand. A capstone project could be the answer for a particular workplace.

Project Themes are chosen to guarantee that each group's projects are complex and innovative. According to the CMO 25 series 2015, the BSIT Capstone should include the following:

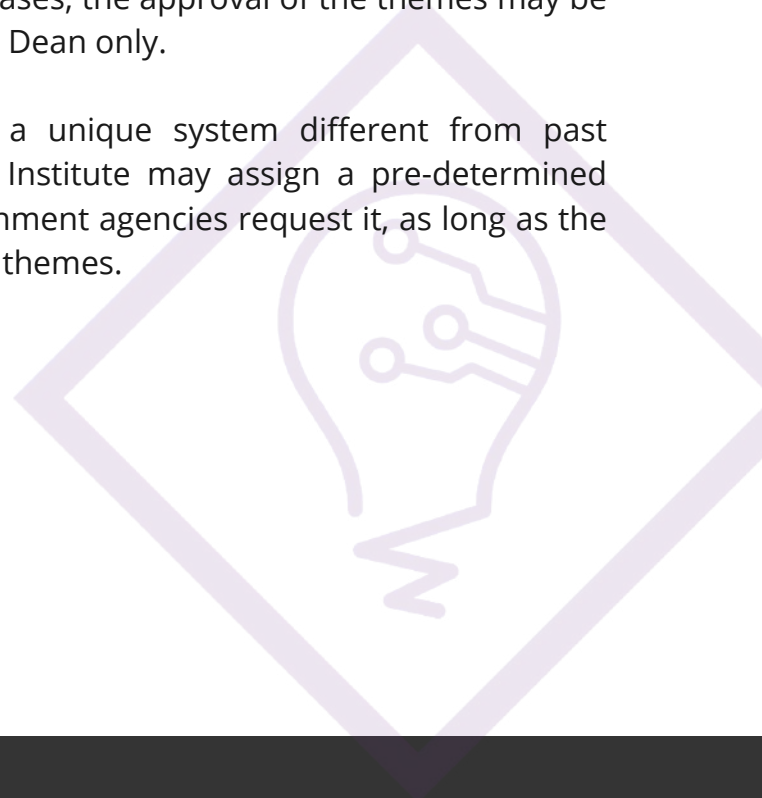
- Software Development
 - Software Customization
 - Information Systems Development for an actual client (with pilot testing)
 - Web Applications Development (with at least alpha testing on live servers)
 - Mobile Computing Systems



- Multimedia Systems
 - Game Development
 - e-Learning Systems
 - Interactive Systems
 - Information Kiosks
- Network Design and Implementation and Server Farm Configuration and Management
- IT Management
 - IT Strategic Plan for sufficiently complex enterprises
 - IT Security Analysis, Planning and Implementation

Nevertheless, to set the level of difficulty of the Capstone projects, the Capstone Project Instructors will prepare and propose themes and present them to the faculty body for approval. However, in exceptional cases, the approval of the themes may be set to the Program Chairperson and Institute Dean only.

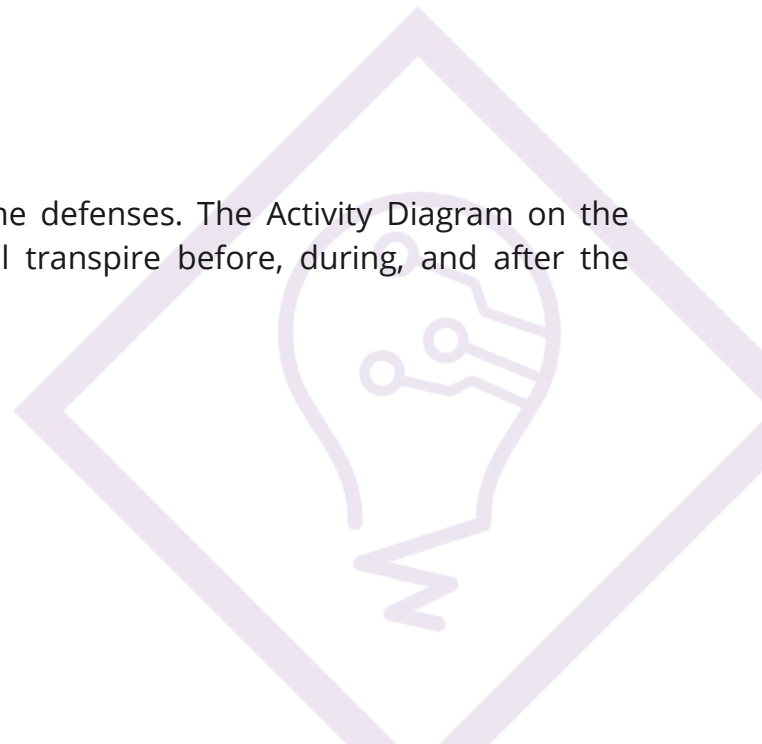
Furthermore, proponents should develop a unique system different from past Capstone Projects. In some instances, the Institute may assign a pre-determined topic to students, especially if partner government agencies request it, as long as the nature of the study remains anchored to the themes.



PROCESS FLOW

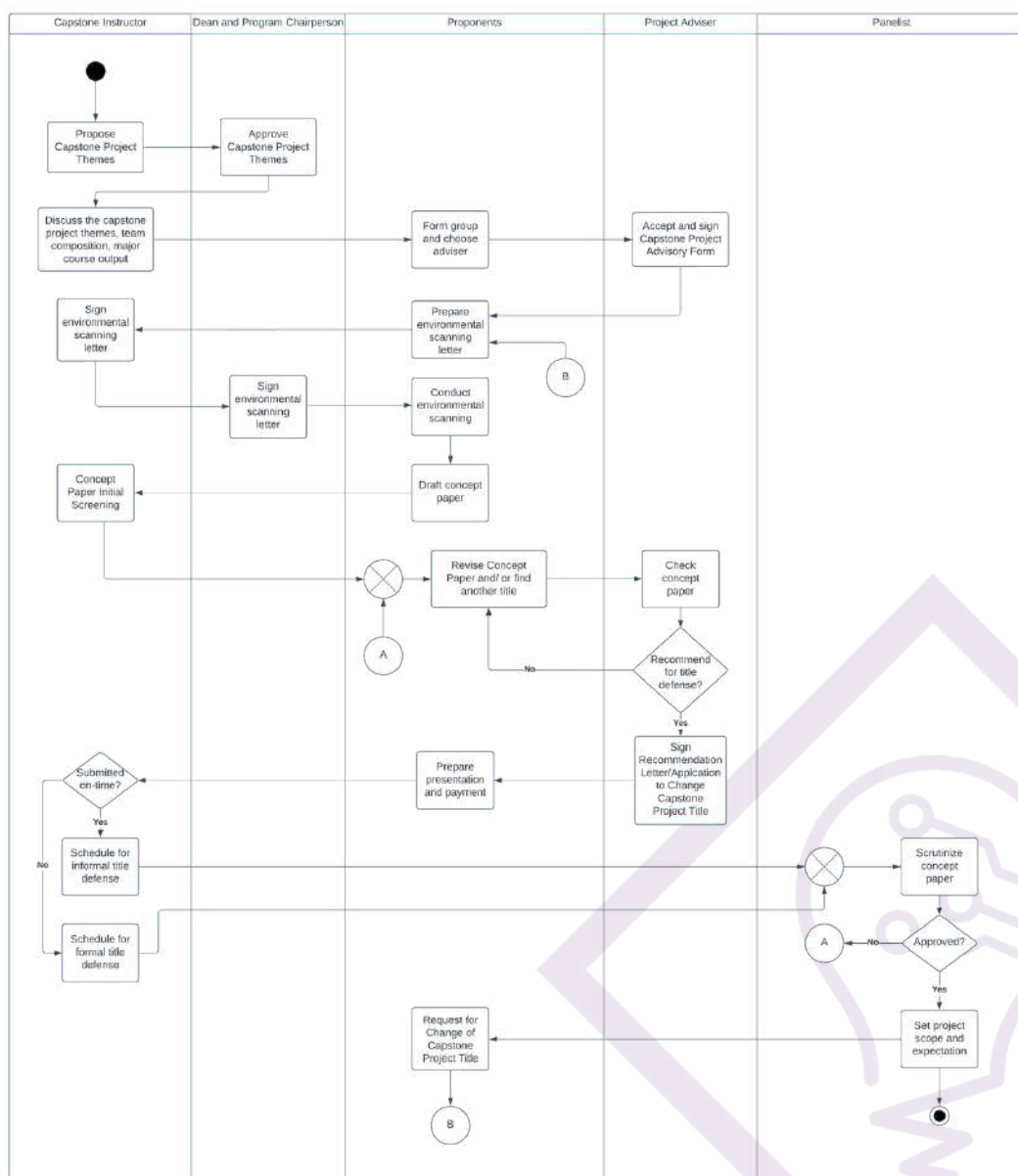


Capstone Project 1 includes the title and outline defenses. The Activity Diagram on the following pages depicts the processes that will transpire before, during, and after the defense.

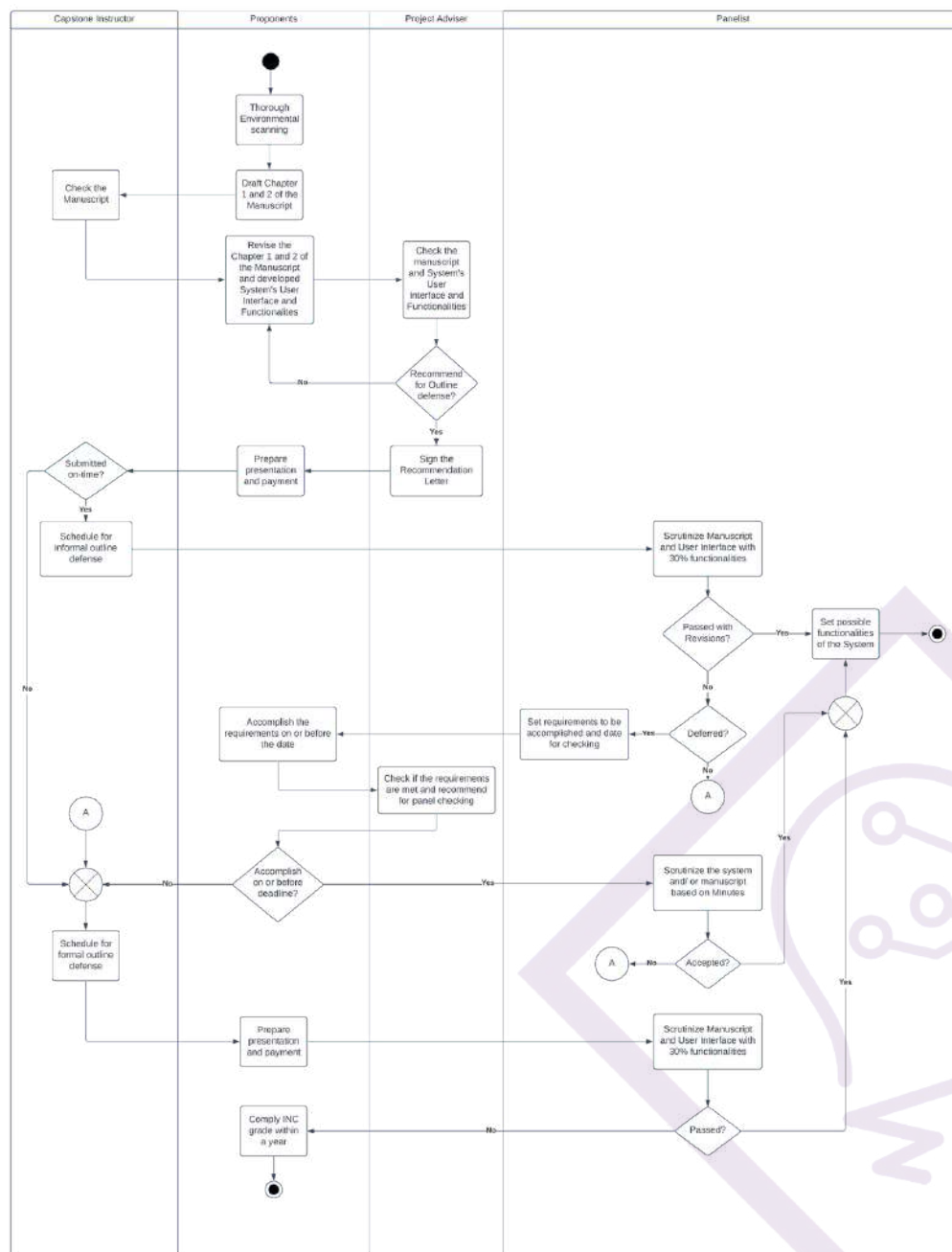




● Title Defense Activity Diagram



Outline Defense Activity Diagram





DOCUMENT CONTENT

● Title

The title is an essential component in every research proposal since it hints to readers whether they will read your paper or not. It should present readers with a clear and concise description of the entire project. The following are some pointers of how to construct a good title (Nair, 2014):

- Contain as few words as possible: many journals limit tags to 12 words
- Be easy to understand
- Describe the contents of the paper accurately and specifically
- Avoid abbreviations, formulas, and jargon
- Not include any verb



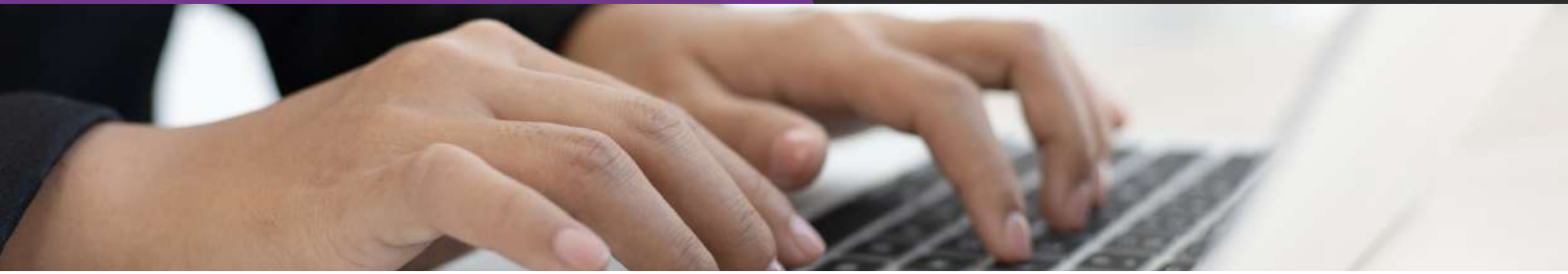
- Not contain low-impact words such as "Some notes on ...," "Observations on ...," "Investigations on ...," "Study of ...," and "Effect of ..."
- Not be flashy as in newspapers (e.g., avoid statements like "Agroforestry can stop deforestation")
- Report the subject of the research rather than the results

Most of the approved titles in DNSC answer the following questions: What, Where, and How. **WHAT** presents the description of the proposed system; **WHERE** details the business profile of the selected organization/business firm that will serve as the project's primary stakeholder; **HOW** (optional) presents to readers the technology/technologies used to develop the system. The following are sample titles of implemented capstone projects in the previous years:

- Management Information System for the Planning Office of the Davao del Norte State College (answers what and where)
- Davao Metro Shuttle Operators Online Bus Dispatching System and Mobile Application: MetroPoint (answers what and where)
- Human Resource Information System (HRIS) for the Davao del Norte State College (answers what and where)
- Requisition and Tracking System for the Supply office of Davao del Norte State College with Product Disposal Forecasting using Linear Regression (answers what, where and how)
- IEC Materials for Office of the Students Services of Davao del Norte State College Using QR codes (answers what, where, and how)

● Chapter 1 - Introduction

The Introduction will set the reader about the general context of the project. It also explains the project's importance, significance, and viability. This chapter also discusses the project's "gap" or problem and how this project addresses that gap. The following sections will further explain the contents and parts needed for this chapter.



Background of the Project

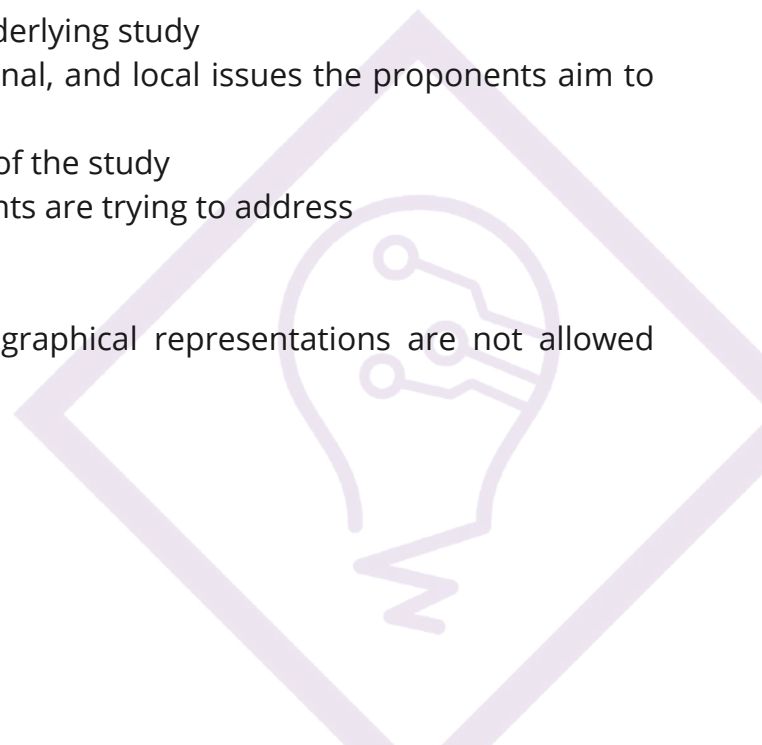
The background of the project presents a literary review of the project. Specifically, the background of the project defines the nature and extent of the problems studied, relates the research to previous work (usually by a brief review of the literature relevant to the problem), explains the objectives of the investigation, and defines any specialized terms or abbreviations to be used in what follows (Nair, 2014).

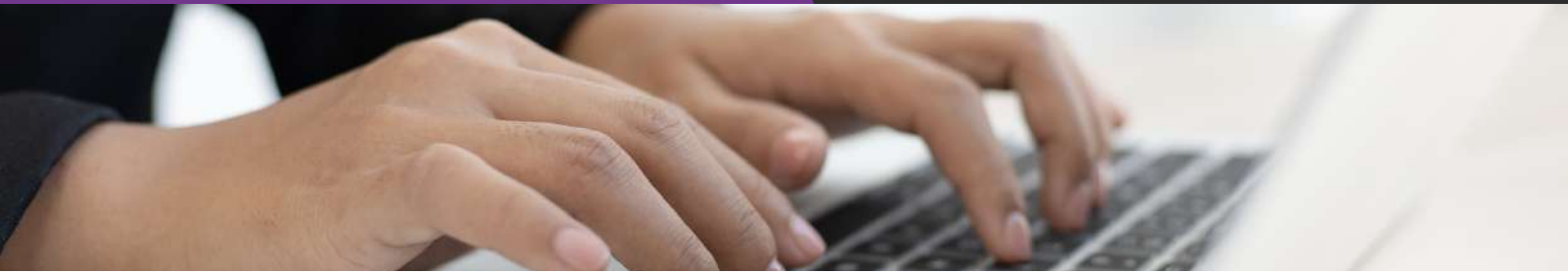
Remember that the project's background logically leads to, and clearly states the hypothesis or principal theme of the paper.

The following are the recommended contents of the background of the project section (note: you may not follow this recommendation, it's up to you how you will organize your contents as long as you present a detailed literature review of the project/system):

- Introductory statement
 - Include context and rationale of the underlying study
 - It should describe the worldwide, national, and local issues the proponents aim to resolve.
- Discuss the SDG and DNSC RDE alignment of the study
- Locale of the study and problems proponents are trying to address
- Overview of the capstone project

Note: Diagrams, graphs, tabular, and other graphical representations are not allowed since this section is narrative in nature.





Objectives of the Study

The study's objectives state what the proponents expect to achieve in general terms and outline the exact details of the project. Each objective should answer all the business problems as cited in the introduction. An objective should start with an action verb and be constructed in SMART format (Specific, Measurable, Attainable/Achievable, Realistic/Relevant, and Time-Oriented). Arrange the specific objectives in an IPO (Input-Process-Output) manner.

Sample (Please be creative and construct your objective):

This study aims to develop an android application called AMP, an Ampalaya disease identifier that provides a treatment and control management guide.

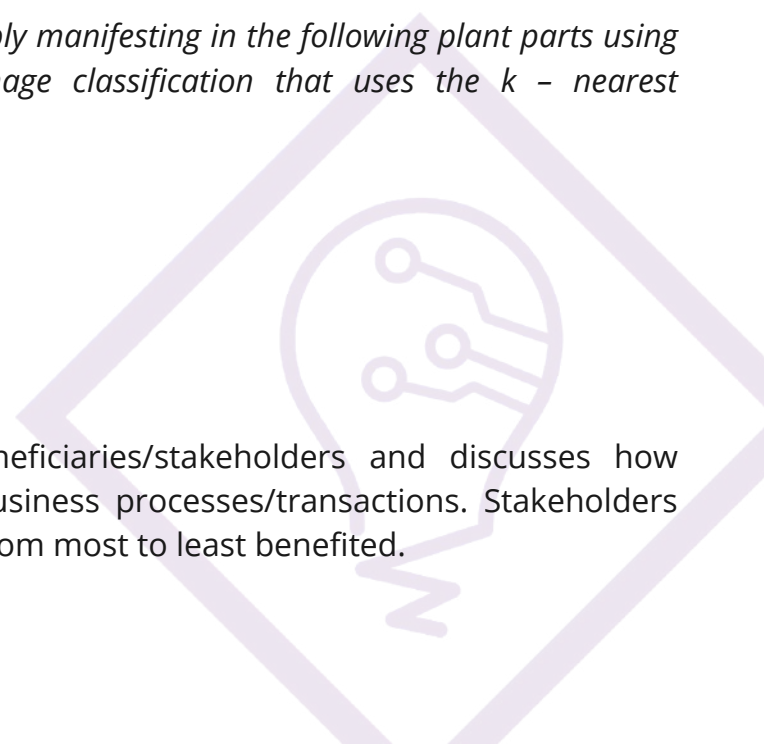
Specifically, the study aims to accomplish the following objectives:

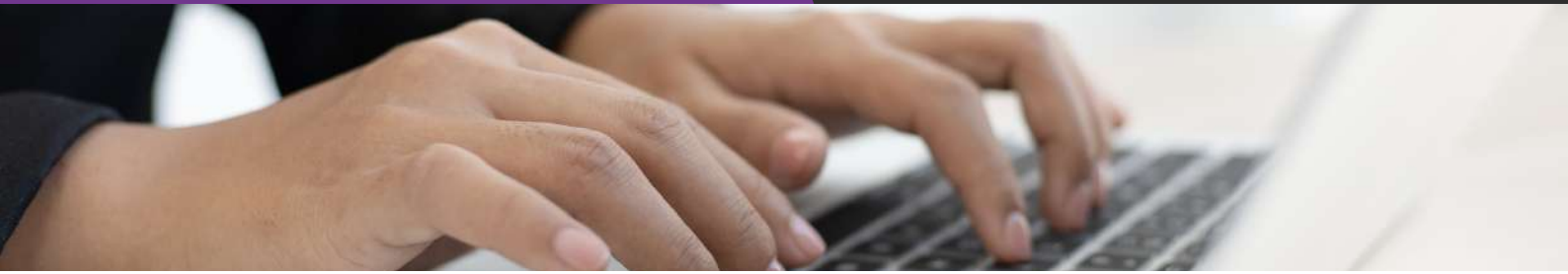
1. Identify diseases of Ampalaya that are visibly manifesting in the following plant parts using image processing techniques, particularly image classification that uses the k – nearest neighbors (KNN) algorithm:

- 1.1. leaves, and*
- 1.2. fruits*

Significance of the Study

This section lists all of the project's beneficiaries/stakeholders and discusses how important the system will enhance their business processes/transactions. Stakeholders should be ranked based on system impact, from most to least benefited.





Scope and Limitation

The scope of the study discusses the study's inclusions and coverage, which include, but are not limited to, (1) the possible location of deployment, (2) potential users of the system, and (3) an overview of the processes involved. The discussion of the scope should be limited to one paragraph.

The limitations are exclusions from the study that were beyond the developers' control, which include but are not limited to (1) processes involving or related to the system that is excluded in the development, (2) technology limitation (i.e., data mining or machine learning features limitation, GIS functionality limitation, game design constraints), (3) platform to be used (web-based, windows-based, online or offline), (4) type of data to be used in the study, and (5) non-functional requirements. Non-functional requirements are descriptions of the system's operation capabilities and limits that improve its functionality. These can include things like speed, security, and dependability.

Review of Related Literature and Works

In the review of related literature and works section, you can expect to find a critical analysis of the existing literature related to your research question. The section typically includes a summary and synthesis of the relevant literature, emphasizing identifying key themes, gaps, and areas of agreement or disagreement. This section includes the discussion of the following:

- Introduction of Related Literature/ Works: This section typically starts with a brief introduction that outlines the literature review's purpose and scope and highlights the research question's importance.
- Overview of existing literature: This section provides a broad overview of the literature related to your problem statement. It may include a historical context, a summary of significant theories or approaches, or a review of key studies or findings.



- **Synthesis of key themes:** In this section, the review will synthesize the existing literature/ works by identifying key themes or patterns that emerge across multiple sources. This may involve summarizing common findings (i.e., the result of the algorithm/s and the parameters used).
- **Identification of gaps:** The review will also identify gaps in the current literature/work and areas where there is a lack of research or conflicting findings. This helps justify the current study's need and sets the stage for the research question. Research gap is usually found under recommendation.
- **Conclusion:** In the conclusion, summarize the key findings you have taken from the literature and emphasize their significance.

Note: Separate the discussion of the Related Literature and Related works. Related literature refers to a critical analysis of the existing literature related to the study's objectives. It is an overview of the relevant literature, identifies key themes, and identifies gaps or areas where further research is needed. In contrast, the related works refer to other systems similar in the proposed system's style, genre, or subject matter.

Provide a tabular comparison of the related works highlighting the similarities and differences of the system functionalities, algorithm, parameters used, and performance metrics. Remember to include your citation. Use IEEE in-line citation.

Definition of Terms

All operational words used in the project are listed in the Definition of Terms section. Terms relevant to the document and technical jargon should be put here, and the definition should explain how the developer utilized the terminologies in the system. All terms must be arranged alphabetically.



● Chapter 2 - Methodology

This chapter must describe how you planned and carried out your project. It would be best to be very comprehensive and specific for this chapter since it will also show how you arrive at your results.

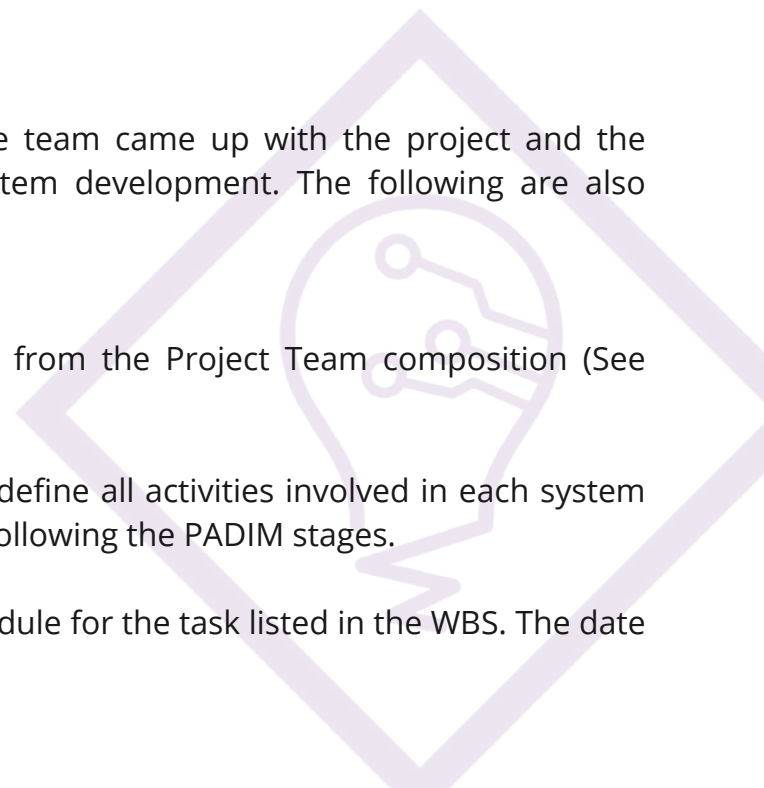
After the introductory statement for this chapter, you also need to discuss your chosen SDLC model that best fits your project. Aside from the model description, you also need to provide a graph that captures that model's respective phases and processes. Each phase or process of the model should also be adequately explained.

This chapter also follows the PADIM or Planning, Analysis, Design, Implementation, and Maintenance structure. To better understand this, the following are the corresponding contents and sections needed for this chapter.

System Planning

This section must thoroughly discuss how the team came up with the project and the description of the planning phase of the system development. The following are also needed for this section:

- Project Team Organization
 - The organizational chart should be based from the Project Team composition (See page 4)
- Work Breakdown Structure (WBS)
 - The WBS should be phased-based and define all activities involved in each system development design phase, preferably following the PADIM stages.
- Gantt chart
 - The Gantt chart depicts the project schedule for the task listed in the WBS. The date format can be month and year.





System Analysis

This section includes the description of the analysis phase of the systems development of the project and must depict the following:

- System Architecture
 - A system architecture explains the process of the system, users, and other system components interacting using the application. It is a conceptual framework that describes the structure and behavior of the proposed system. Moreover, this section describes the system components, including hardware, software, and data components. It includes diagrams or other visual representations to help illustrate the architecture.
- Functional and non-functional requirements
 - This section should discuss the system's expected system functionalities (functional requirements) and expected quality attributes (non-functional requirements). Provide a brief discussion and numbered list form of the two requirements.
- Use Case Diagram
 - A Use Case Diagram highlights the details of system users' (also known as actors) and interactions with the proposed system.
- Context Flow Diagram (Level 0 Data Flow Diagram)
 - The Context Flow Diagram in the current system depicts the flow of business actions (logical). It specifies the business events and the data each event requires and produces. It also shows how the system interacts with external entities (system, organization groups, external data stores, etc.). However, the physical data flow will be utilized if the theme falls under game development.
- Level 1 Data Flow Diagram
 - The first explosion of the Level 0 DFD is Level 1. It shows the Level 1 DFD in greater detail and depicts the system's basic modules and data flow. The system is broken down into sub-processes in this diagram. Note that the entities and data coming in and out from the entity in the Level 0 DFD should retain in Level 1 DFD.

Note: For logical diagrams, Level 0 and Level 1 DFD, use the Gane and Sarson method.



System Design

This section must depict the description and details of the project's systems development design phase. The following are the other corresponding contents for this section:

- Entity Relationship Diagram (ERD)
 - An ERD depicts the relationships between database entity sets. An object, or a data component, is an entity in this context. A physical data model should depict a relational database's actual architecture plan.
- JSON Schema Diagram
 - JSON Schema is a vocabulary to annotate and validate JSON documents. It defines how a JSON should be structured, making it easy to ensure that a JSON is formatted correctly, and it is useful for automated testing and validating.
- Data Dictionary
 - The data dictionary of the system should be displayed in this section. Table <table #> shows the list of all tables, and Table <table #>.1, Table <table #>.2, and so on, should be used for the succeeding tables.
- Technologies, Concepts, and Theories
 - This section deeply discusses the model/ algorithm/ emerging technologies shown in the system architecture. It includes the discussion of the following but is not limited to, depending on the study's objectives and the process involved in the model.
 - Data collection
 - Data pre-processing
 - Feature extraction
 - Classification
 - This part should also include a list of the technologies required to develop the system. A definition of the technology, a snapshot of the Interface as the proponents used it, and a discussion of how the technology was utilized should all be included in the manuscript.

Note: For a system with SQL based database, the JSON Schema Diagram shall be omitted; however, systems with NoSQL, ERD shall be excluded.



System Development and Testing

This section includes a detailed description of the tests performed to ensure the system functions correctly and meets the requirements specified in the system design phase. The system testing phase typically includes the following components:

- Result of the testing the model including the performance metrics
- Summary of the findings (provide tabular representation)

This section also includes the description of the deployment and implementation phase of the systems development of the project. It must also have the **Systems Test Plan** and **Systems Implementation Plan**. A System Test Plan lays out the entire process of validating the testing process. At the same time, a System Implementation Plan describes all the activities associated with implementing and evaluating an information system.

System Maintenance

This section includes the description of the maintenance phase of the systems development of the project. It must also have the **Systems Security Plan** and **Systems Maintenance Plan**.

Note: Because the project has still not reached the System Implementation and System Maintenance stages, these parts should be written in the future tense.

● References

The last page of the manuscript is the References page, which should be written in IEEE style. It contains a list of all the sources you've used in your manuscript, making it easy for readers to locate what you've cited. The IEEE format should be used for in-line citations in the manuscript. You may use visit this link for reference -

<https://ieeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf>.

The numerical arrangement should also be based on the chronological appearance of the respective reference.



System Testing and Implementation

This section includes a detailed description of the tests performed to ensure the system functions correctly and meets the requirements specified in the system design phase. The system testing phase typically includes the following components:

- Result of the testing the model including the performance metrics
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<https://ieeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf>.

The numerical arrangement should also be based on the chronological appearance of the respective reference.



DOCUMENT FORMAT

● Styling Format

HEADING

Alignment: Centered; Font-weight: Bold; Font-case: Uppercase

SUB-HEADING

Alignment: Left-align; Font-weight: Bold

FONT

Font style: Arial; Font size: 12; Line spacing: 1.5

CONTENT / TEXT

Alignment: Justified; Indention: 0.5 (first line)

MARGINS

Left: 1.5 Inches"; Top: 1 Inch"; Right: 1 Inch"; Bottom: 1 Inch"

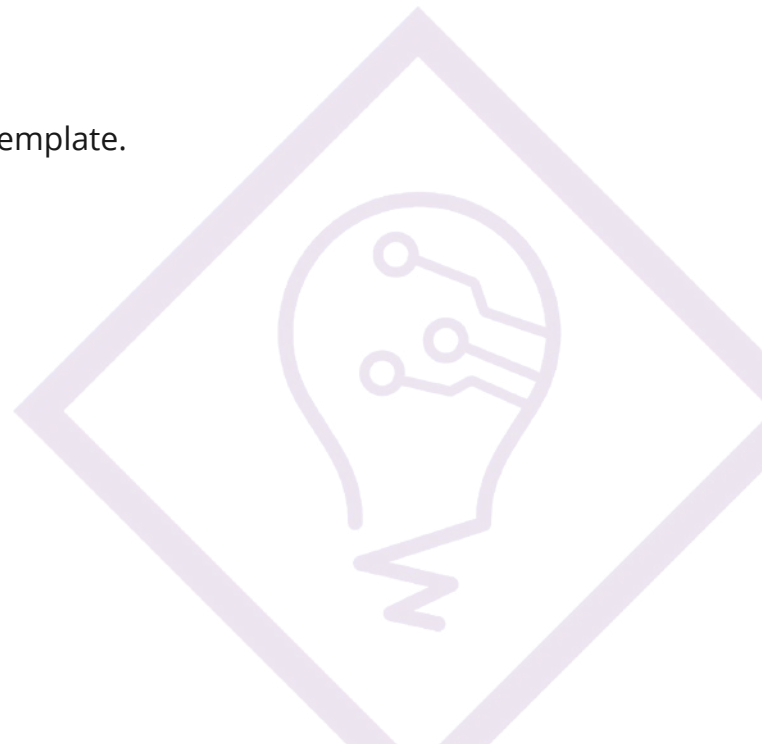
REFERENCES

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DOCUMENT TEMPLATE

This portion shows the manuscript's document template.



● Cover Page

<PROJECT TITLE>



DAVAO DEL NORTE STATE COLLEGE

Institute of Computing
New Visayas, Panabo City

A Capstone Project Presented by:

<Name of the proponents in alphabetical order – Format: First Name MI. Last name>

Month Year

**Left-Margin
1.5 Inches**

● Sample Content Arrangement

CHAPTER 1

INTRODUCTION

Background of the Project

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Quisque lacinia mi est, nec tristique [2] [3] metus maximus non. Donec quis sem a purus cursus ornare. Suspendisse vel tortor et quam rhoncus sollicitudin. Quisque id est ante. Sed neque tortor, eleifend et lacus eget, ultrices vulputate diam. Nulla venenatis sit amet metus vel tempor. Integer vitae nunc ultrices, rhoncus arcu quis, luctus ante. Interdum et malesuada fames ac ante ipsum primis in faucibus. Donec ultrices semper pharetra. Fusce eget tellus semper, blandit risus at, aliquam mauris. Proin malesuada libero sit amet nunc pretium, et congue leo molestie. Suspendisse mi erat, congue id venenatis sit amet, aliquam vel nisi. In ultricies commodo volutpat. Fusce a pellentesque purus. Proin eget porttitor ex.

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Objectives of the Study

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1. In hac habitasse platea dictumst;
2. Curabitur ut urna ligul;
3. Generate the following reports:
 - 3.1. Phasellus nec massa quis;
 - 3.2. Maecenas gravida convallis; and
 - 3.3. Ut vitae laoreet lorem.

Significance of the Study

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Future Developers. Maecenas gravida convallis mauris id tincidunt.

Scope and Limitation

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Review of Related Literature and Works

Related Literature

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Related Works

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Definition of Terms

Aliquam. Aliquam pellentesque rutrum congue.

Curabitur. Curabitur ut urna ligula. Nulla in mauris libero.

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CHAPTER 2

METHODOLOGY

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Figure 1. Title of Figure



System Planning

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Project Team Organization

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Work Breakdown Structure (WBS)

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Gantt Chart

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System Analysis

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officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae.

System Architecture

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Functional and Non-Functional Requirements

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Use Case Diagram

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Context Flow Diagram

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Level 1 Data Flow Diagram

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System Design

Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

Entity Relationship Diagram (ERD)

Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae.

JSON Schema Diagram

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Data Dictionary

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Table 1.1 Name of Table

Attributes	Data Type	Key Type	Description

Technologies, Concepts, and Theories

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System Testing and Implementation

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nulla pariatur.

System Maintenance

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veritatis et quasi architecto beatae vitae dicta sunt explicabo.



REFERENCES

- [1] Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
- [2] Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.
- [3] Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
- [4] Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.
- [5] Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium.
- [6] totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.
- [7] Nulla venenatis sit amet metus vel tempor. Integer vitae nunc ultrices, rhoncus arcu quis, luctus ante. Interdum et malesuada fames ac ante ipsum primis in faucibus.
- [8] Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias.
- [9] consequatur aut perferendis doloribus asperiores repellat.



DEFENSE REQUIREMENTS

This section discusses the requirements for both Title and Outline Defense.





Title Defense Requirements

The Capstone Project Instructor must receive the enumerated prerequisites before the proponents can be scheduled for Title Defense.

- Copy of Approved Recommendation Letter signed by the Adviser
- Soft Copy of Concept Paper of the three (3) proposed projects (see the format in Concept Paper under Document Template)
- Environmental Scanning Documents/ Letters duly received by the person or representative address in the letter
- Proof of Payment

Outline Defense Requirements

The Capstone Project Instructor must receive the enumerated prerequisites before the proponents can be scheduled for Outline Defense.

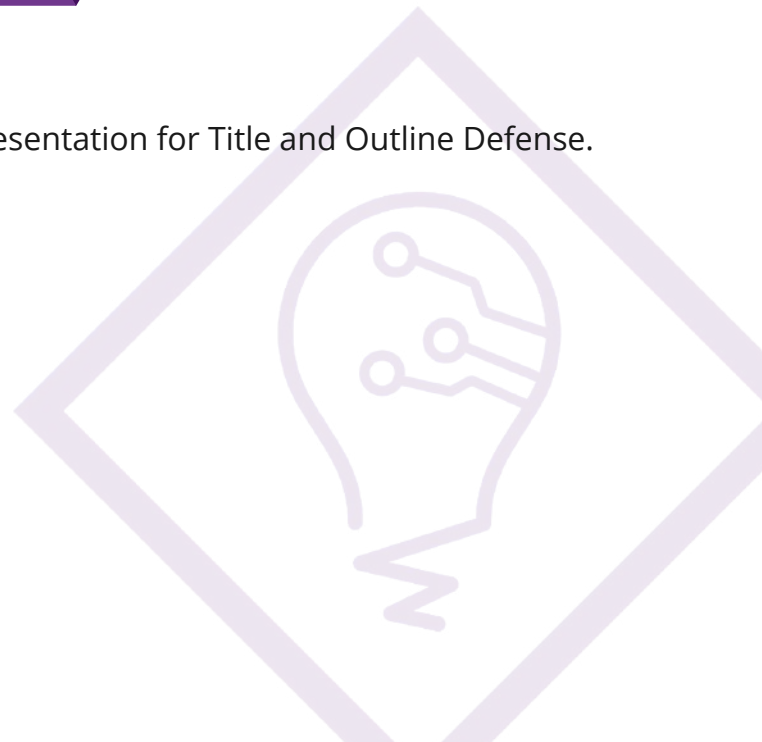
- Copy of Approved Recommendation Letter signed by the Adviser
- Soft Copy of Manuscript (Chapter 1 and 2)
- System with 100% User Interface and 30% Functionalities
 - 30% System functionalities is defined during the BSIT Faculty meeting levelling
- Copy of Acceptance and Rejection Letters duly received by the person or representative address in the letter
- Proof of Payment

Note: For both title and outline defense, failure to submit either one of the requirements before the deadline means no schedule for the defense.



PRESENTATION CONTENT AND FORMAT

This section discusses the expected students' presentation for Title and Outline Defense.





Title Defense

During the title defense, the students are expected to present their proposed Capstone Project study's three (3) concept papers to the panelists. The following is the outline of the presentation during the title defense:

Outline	Description
Introduction	The proponents must be able to introduce their names to the panelists.
Concept of the three (3) Capstone Project study	The concept of the Capstone Project study includes the following: • Project Title • Rationale and brief background of the problem statement, locale and overview of the project. The presentation should be based from the Problem Tree Analysis, Solution Tree Analysis and MCCA Matrix • Objectives of the Study • Scope and Limitation • Comparative Table with existing system and proposed system (if possible)
Question and Answer	The proponents are expected to answer the question thrown by the panelists relevant to the proposed Capstone Project. The panelists will ask the proponents on the overview of the study (including the current process flow of the organization relative to the study), novelty or main contribution of the proponents , objectives of the study and possible methodologies.

Outline Defense

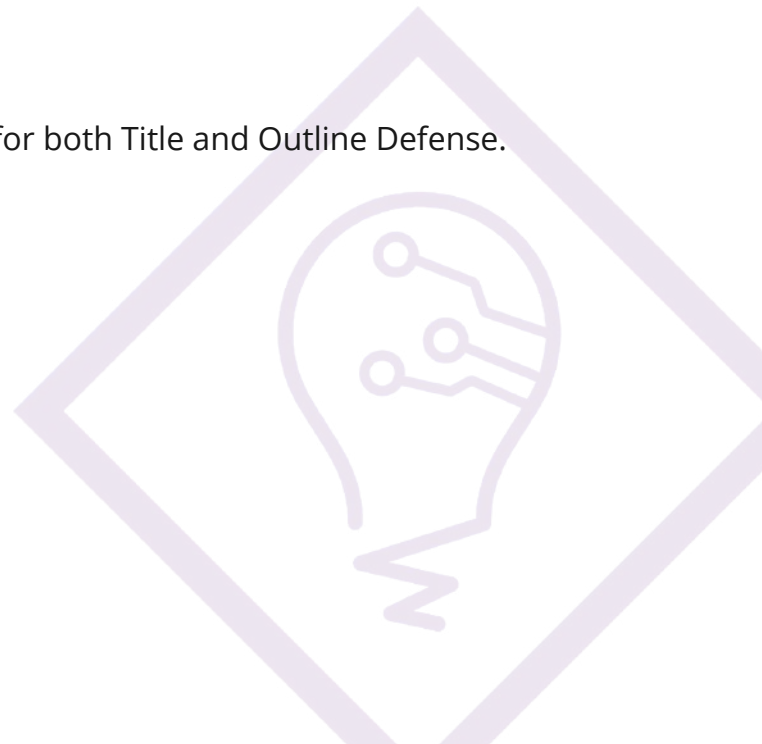
During the outline defense, the students are expected to present their approved Capstone Project Title and System prototype . The following is the outline of the presentation during the outline defense:

Outline	Description
Introduction	The proponents must be able to introduce their names to the panelists.
Executive Summary	The executive summary is a brief background of the capstone project. This details the following: • Project Title • Rationale and brief background of the problem statement • Objectives of the Study • Comparative presentation of the Capstone project and existing system • Brief discussion of the methodologies (Functional and non-functional requirements, Conceptual framework, use case diagram, CFD and Level 1 DFD, ERD/ JSON Schema Diagram, Technologies used)
System prototype walkthrough and Simulation presentation	The proponents will walkthrough the panelists to their system prototype based on the functional requirements of the system. Simulation of the model/ sensors will follow.
Question and Answer	The proponents are expected to answer the question thrown by the panelists relevant to the system prototype and model simulation. The panelists will ask the proponents about the system's flow, the algorithm used in the model (including parameters) or sensors used, and other pertinent questions relative to the title.



MINUTES TEMPLATE

This portion shows the template of the minutes for both Title and Outline Defense.





DAVAO DEL NORTE STATE COLLEGE

Institute of Computing

MINUTES OF THE <Name of the Defense>DEFENSE

Date: _____ Time Started: _____ Time Ended: _____

Title of the Study: _____

Proponents:

1. _____ 4. _____
2. _____ 5. _____
3. _____

Panelists:

Chairman: _____ Signature: _____

Members:

1. _____ Signature: _____
2. _____ Signature: _____

DOCUMENT COMMENTS:

Panelist 1:

1. _____
2. _____

Panelist 2:

1. _____
2. _____

Panelist 3:

1. _____
2. _____

SYSTEM COMMENTS: (if Outline Defense)

Panelist 1:

1. _____
2. _____

Panelist 2:

1. _____
2. _____

Panelist 3:

1. _____
2. _____

Verdict: ☐ Passed with Revisions ☐ Deferred until _____ ☐ Failed

The following are the requirements for the students with Deferred verdicts.

1. _____
2. _____

Prepare by:

Adviser

Note: The header and footer may change depending on the ISO update.
Download the .docx file [here](#).



Davao del Norte State College

INSTITUTE OF COMPUTING

Bachelor of Science in Information Technology



End of Capstone Guideline

THANK YOU

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